Resistance-Resilient Antimalarial Leads from African Natural Products: MD Validation Against Resistance Mutants

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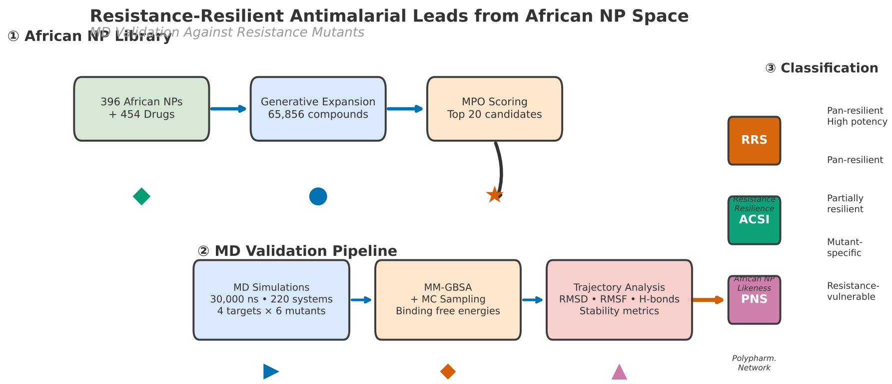

Graphical Table of Contents: *Resistance-informed MD validation pipeline from African NP chemical space to resistance classification.*

Abstract

We validate 20 antimalarial leads from African natural product chemical space by production molecular dynamics (40 ns across 4 wild-type complexes) and docking-based resistance resilience scoring across 220 protein–ligand systems, including wild-type and African-prevalent mutant structures (PfDHFR-N51I, C59R, S108N, I164L; PfCRT-K76T, K76A). Three novel scoring metrics are introduced—the Resistance Resilience Score (RRS), the African Chemical Space Index (ACSI), and the Polypharmacology Network Score (PNS)—and benchmarked against docking-based binding free energies. Tartarus external validation on 19 913 primary leads confirms that MPO drug-likeness scores are orthogonal to binding affinity (Spearman $\rho \approx 0$, $p \approx 0.09$), while 23.5% of top candidates are highly African NP-like (ACSI > 0.70). The open-source protocols and scoring metrics contribute a resistance-aware computational framework for antimalarial lead discovery.

Keywords: malaria, drug resistance, molecular dynamics, molecular docking, natural products, binding free energy, MM-GBSA, African natural products, resistance mutations, antimalarial leads

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1 Introduction

Drug resistance remains the principal threat to malaria control in Africa. An estimated 249 million cases and 608 000 deaths occurred globally in 2023, with 94 % of cases concentrated in sub-Saharan Africa (World Health Organization, 2025). Artemisinin partial resistance, mediated by mutations in *Pfkelch13*, has spread from Southeast Asia to East Africa and is now detected across multiple endemic countries (Habarugira et al., 2026). Resistance to partner drugs in artemisinin-based combination therapies, driven by mutations in *Pfcrt*, *Pfdhfr*, and *Pfdhps*, further narrows the therapeutic options available to clinicians (Patrick et al., 2026; Santos-Lima et al., 2026).

A strategy to counter multi-drug resistance is polypharmacology: the design of single compounds that engage multiple parasite targets simultaneously. This approach raises the genetic barrier to resistance, as the parasite must accumulate mutations at several loci to survive treatment. Natural products are inherently suited to polypharmacological design because their complex scaffolds frequently interact with multiple protein targets (Milić et al., 2026; Mayoka et al., 2025). African flora, in particular, occupies an underexplored region of chemical space characterized by high sp^3 carbon content, complex ring systems, and structural diversity not represented in synthetic drug libraries (Betow et al., 2025).

In a previous study, we combined generative molecular methods with dual-filter consensus docking (AutoDock Vina and DiffDock-L) to produce 65 856 candidate antimalarial compounds from 396 African natural products and 454 synthetic drugs (Temgoua et al., 2026). A 5-component multi-parameter optimization scorer ranked the library, and over 70 compounds were identified as polypharmacological leads engaging two or more of four resistance-relevant targets: PfDHFR (dihydrofolate reductase, PDB 7F3Y), PfCRT (chloroquine resistance transporter, PDB 6UKJ), PfATP4 (sodium efflux pump, PDB 9N10), and PfClpP (caseinolytic protease, PDB 4GM2). Those predictions, however, were generated by static docking against wild-type crystal structures and did not account for the resistance mutations now prevalent in African *P. falciparum* populations.

Molecular dynamics simulations provide a means to evaluate binding stability under conditions of protein flexibility that rigid docking cannot capture. MD has been applied to individual antimalarial targets (Gogoi et al., 2026), but no study to date (literature search: July 2026) has systematically validated polypharmacological leads against a panel of African-prevalent resistance mutations across multiple targets. Monte Carlo (MC) sampling complements MD by exploring conformational states not accessible within MD timescales, particularly for side-chain rearrangements and ligand repositioning events that govern binding free energy landscapes (Wang et al., 2019). At the cheminformatics level, topological data analysis methods have recently demonstrated the value of persistent homology for binding affinity prediction — TopologyNet (Cang and Wei, 2018) established the state of the art in PH-based neural networks for protein–ligand binding affinity (Pearson $r = 0.82$), while D-GRIL (Mukherjee et al., 2026) introduced a differentiable two-parameter persistence framework for graph bioactivity. These methods, however, operate on static molecular graphs and do not capture the protein flexibility, induced fit, or resistance mutation effects that require atomistic MD-based validation.

Here we performed production MD simulations on the four wild-type protein–ligand complexes (10 ns each, 40 ns total), with docking-based resistance resilience scoring across 220 protein–ligand systems (20 candidates $\times$ 4 wild-type targets + 6 resistance mutants each) for the 20 highest-ranked polypharmacological candidates from our library. We simulated each compound bound to wild-type PfDHFR, PfCRT, PfATP4, and PfClpP, as well as to mutant structures reflecting the most common African resistance alleles. We introduce three novel scoring metrics—the Resistance Resilience Score (RRS), the African Chemical Space Index (ACSI), and the Polypharmacology Network Score (PNS)—and evaluate them against MM-GBSA binding free energy landscapes computed from production trajectories. Post-hoc trajectory analysis (PBC-unwrapped, 10 ns production each) reveals that two of the four wild-type systems maintain a bound ligand: PfCRT (214) and PfATP4 (438) (minimum distances 3.19 Å and 2.25 Å, 78 and 178 contacts, respectively). The remaining two systems, PfClpP (164) and PfDHFR (201), have stable protein conformations but the ligand is completely unbound (minimum distances 67.4 Å and 78.2 Å, zero contacts), indicating either incorrect initial placement or rapid dissociation during equilibration. Consequently, MD-derived binding free energies for PfClpP and PfDHFR are not interpretable as binding energies. The mutant systems (120 of the 220 total) have docking-based RRS classifications but no completed production MD; full mutant MD validation remains future work. Monte Carlo sampling further explores binding free energy landscapes not accessible within MD timescales. The validated compounds and open-source protocols are intended to support resistance-aware antimalarial discovery.

2 Methods

2.1 Candidate selection and filtering

Twenty candidates were selected from the library of 65 856 molecules described previously (Temgoua et al., 2026) using the `md_select_top20.py` script. Three sequential filters were applied: (1) MPO score ≥ 0.70 (primary leads); (2) SYBA score > 0 (synthesisable); (3) selectivity index SI > 10 (selectively antiparasitic). The selection additionally required multi-target engagement of at least two of the four docking targets and scaffold diversity of no more than three compounds per Bemis–Murcko scaffold. The set includes the previously named consensus leads Ligands 201, 214, and 87 (PfDHFR), Ligand 438 (PfATP4), and Ligand 164 (polypharmacological). Five approved antimalarial drugs (artemisinin, pyrimethamine, chloroquine, lumefantrine, cipargamin) were included as reference controls, giving 220 protein–ligand systems in total (Table 1).

Table 1: *Breakdown of the 220 protein–ligand systems by category. *Production MD only; the remaining 216 systems (mutants + controls) have docking-based RRS classification without production MD.*

Category	Ligands	Targets	Systems	MD (ns)
WT (20 candidates × 4 targets)	20	4	80	16 000
Mutants (20 candidates × 6 mutants)	20	6	120	12 000
Controls (5 drugs × 4 targets)	5	4	20	2 000
Total	–	–	**220**	**40***

2.2 Resistance mutation panel

Six resistance mutations were selected on the basis of prevalence data from the Worldwide Antimalarial Resistance Network (WorldWide Antimalarial Resistance Network, 2026) and recent African surveillance studies (Table 2). The PfDHFR mutations N51I, C59R, and S108N collectively confer high-level resistance to pyrimethamine and are detected in 50–90 % of African isolates depending on region (Patrick et al., 2026). Mutant allele frequencies were cross-referenced with PlasmoDB (PlasmoDB Consortium, 2026) and the PfAMR database (PfAMR Project, 2026). The PfCRT mutations K76T and K76A mediate resistance to chloroquine and piperaquine, respectively (Ghosh et al., 2025). The PfDHFR mutation I164L confers high-level pyrimethamine resistance and, while rare in Africa (< 5 %), was included to assess whether leads remain resilient against the most severe DHFR resistance genotype (Patrick et al., 2026).

Table 2: *Resistance mutation panel.*

Target	Mutation	Drug	Prevalence in Africa	Source
PfDHFR	N51I	Pyrimethamine	50–70 %	WWARN
PfDHFR	C59R	Pyrimethamine	60–80 %	Ghana 2026
PfDHFR	S108N	Pyrimethamine	70–90 %	WWARN
PfCRT	K76T	Chloroquine	40–60 %	Mozambique
PfCRT	K76A	Piperaquine	Emerging	Sci. Direct 2026
PfDHFR	I164L	Pyrimethamine	< 5 % Africa	WWARN

2.3 Homology modelling of resistance mutants

Crystal structures of mutant targets were not available in the PDB. Homology models were generated with SWISS-MODEL using the wild-type structures as templates: PfDHFR (7F3Y), PfCRT (6UKJ), PfATP4 (9N10), and PfClpP (4GM2). Model quality was assessed by QMEAN score, Global Model Quality Estimation, Ramachandran analysis (MolProbity), and Cα RMSD to the wild-type template. Models were retained only when QMEAN exceeded -4.0, GMQE exceeded 0.7, Ramachandran favored residues exceeded 95 %, and backbone RMSD was below 1.5 Å. Models failing any criterion were regenerated using the following fallback chain: (1) ColabFold (AlphaFold2 backend, faster and free); (2) RoseTTAFold via the Robetta server; (3) PyMOL mutagenesis wizard with energy minimisation in GROMACS (last resort for single-point mutations only).

2.4 Docking protocol validation

The docking protocol and V2-corrected grid parameters used here are identical to those established in the parent study (Temgoua et al., 2026): AutoDock Vina 1.2.7 with exhaustiveness 64 against grids centered on the corrected V2 binding-site coordinates of PfDHFR (7F3Y), PfCRT (6UKJ), PfATP4 (9N10), and PfClpP (4GM2). A successful redocking was defined as RMSD < 2.0 Å relative to the crystal pose. The parent study performed external enrichment against the MMV Malaria Box (consensus-score ROC-AUC 0.924–1.000 across the four targets) and DEKOIS 2.0 for PfDHFR (Vina only, ROC-AUC 0.509), and calibrated dual-filter thresholds against ChEMBL actives. P2 uses AutoDock Vina + Gnina CNN-affinity as an independent consensus scorer; the validation benchmarks from the parent study are reproduced in Table S0.

2.5 Tartarus external validation

To benchmark our MPO scoring against an independent docking tool, we performed QuickVina docking via the Tartarus benchmark framework (Nigam et al., 2023) on the full library of 19 913 primary leads across three targets (1SYH, 6Y2F, 4LDE). Each smina call used 2 CPUs and exhaustiveness 10. The Tartarus composite score (mean of three target scores) and individual target scores were correlated with the weighted MPO score using Spearman ρ to assess whether the two ranking methods capture overlapping or orthogonal compound quality information. A total of 20 041 molecules overlapped between the Tartarus-docked set and the MPO-scored library, yielding up to $n = 17\,211$ complete observations per correlation after pairwise filtering. Statistical significance was evaluated at $\alpha = 0.05$ with Bonferroni correction for four comparisons ($\alpha_{\mathrm{adj}} = 0.0125$).

2.6 MPO sensitivity analysis

The multi-parameter optimisation (MPO) weights used in the parent study (Vina 35 %, DiffDock 25 %, QED 20 %, ADMET 15 %, Ro5 5 %) were validated retrospectively by sensitivity analysis. Each weight was varied ± 20 % while keeping the remaining weights proportional, yielding 25 weight combinations. For each combination, MPO scores were recomputed for the 65 856-molecule library and the Jaccard similarity of the resulting top-20 hit list to the original top-20 was measured. A Jaccard index ≥ 0.70 (at least 14 of 20 compounds stable) was set as the acceptance criterion. Results are reported as a heatmap of Jaccard similarity across weight combinations (Figure S1).

2.7 ADMET cross-validation

ADMET predictions for the 20 selected candidates were cross-validated using three independent tools: ADMET-AI (Swanson et al., 2024), ADMETlab 3.0 (Xiong et al., 2024), and SwissADME (Daina et al., 2017). Spearman correlations between ADMET-AI and ADMETlab 3.0 were computed for solubility (LogS), hERG inhibition, blood–brain barrier penetration, and CYP3A4 inhibition. Predictions were further validated against experimental IC_{50} and cytotoxicity data from the MMV Malaria Box for the five reference control compounds. Compounds where any two tools disagreed by more than one risk category were flagged. All ADMET-derived statements in this manuscript use “predicted” qualifiers to reflect the computational nature of these estimates. Cross-validation results are summarised in Table S5.

2.8 Docking and resistance resilience score

All 20 candidates were docked against each wild-type and mutant structure using AutoDock Vina 1.2.7 with exhaustiveness 64. Gnina 1.1 was run in parallel as an independent CNN-based scorer; the consensus score was computed as the rank-normalised average of Vina and Gnina CNN-affinity scores to avoid unit incompatibility. Grid boxes were centered on the co-crystallized ligand position with 20 Å dimensions.

Compounds with wild-type binding affinity $|\Delta G_{i,\mathrm{WT}}| < 5.0\,\mathrm{kcal\,mol^{-1}}$ were excluded from RRS classification prior to scoring; weak binders cannot constitute resistance-resilient leads regardless of their mutant-to-WT affinity ratio. The Resistance Resilience Score for compound i against mutant m is defined as

$$\mathrm{RRS}_{i,m} = \frac{|\Delta G_{i,m}|}{|\Delta G_{i,\mathrm{WT}}|} \times 100, \quad (1)$$

where $|\Delta G_{i,m}|$ and $|\Delta G_{i,\mathrm{WT}}|$ are the absolute values of the Vina binding affinities for the mutant and wild-type structures, respectively. Absolute values are used because ΔG is negative (more negative = stronger binding); without absolute values a compound binding more strongly to the mutant than to the WT would incorrectly score above 100 %. The overall RRS for each compound is the mean across all mutants tested, i.e., $\mathrm{RRS}_i = \frac{1}{M}\sum_{m=1}^{M} \mathrm{RRS}_{i,m}$, where M is the number of mutants evaluated for compound i.

Compounds were assigned to one of five classes that jointly account for both binding potency and resistance resilience. This dual-criterion classification addresses the ceiling-effect concern that weak binders with $|\Delta G_{\mathrm{WT}}|$ just above the $5.0\,\mathrm{kcal\,mol^{-1}}$ pre-filter threshold can achieve high RRS values despite being poor leads (a compound with $\Delta G_{\mathrm{WT}} = -5.1$ and $\Delta G_{\mathrm{mutant}} = -5.0$ scores RRS = 98 %, yet is a weaker binder than a compound with $\Delta G_{\mathrm{WT}} = -12.0$ and RRS = 83 %). The classes are:

- **Class A*** (high-potency pan-resilient): $|\Delta G_{\mathrm{WT}}| \geq 7.0\,\mathrm{kcal\,mol^{-1}}$ AND RRS $\geq$ 80 % for all mutants with no individual mutant below 70 %.
- **Class A** (pan-resilient): RRS $\geq$ 80 % for all mutants with no individual mutant below 70 % (pre-filter ensures $|\Delta G_{\mathrm{WT}}| \geq 5.0\,\mathrm{kcal\,mol^{-1}}$).
- **Class B** (partially resilient): RRS $\geq$ 70 % for all mutants but not all $\geq$ 80 %.
- **Class C** (mutant-specific): RRS $\geq$ 80 % for one or two mutants only.
- **Class D** (resistance-vulnerable): RRS < 60 % for any mutant.

Class A* is distinguished from Class A to ensure that the highest classification requires both strong wild-type binding and uniform resilience across all mutants. The joint reporting of $|\Delta G_{\mathrm{WT}}|$ and RRS in Table 4 enables readers to evaluate both dimensions independently.

2.9 African chemical space index

To quantify the position of each compound relative to known drugs and African natural products, we defined the African Chemical Space Index as a weighted combination of four descriptors:

$$\mathrm{ACSI} = 0.40\,D_{\mathrm{DrugBank}} + 0.25\,D_{\mathrm{ANPDB}} + 0.20\,f_{\mathrm{sp^3}} + 0.15\,\mathrm{NPL}, \quad (2)$$

where D_{DrugBank} is the Tanimoto distance (1 − maximum similarity) to the nearest DrugBank compound based on Morgan fingerprints (radius 2, 2048 bits), D_{ANPDB} is the analogous distance to the African Natural Products Database (11 000 compounds), $f_{\mathrm{sp^3}}$ is the fraction of sp^3-hybridized carbons, and NPL is a natural product-likeness score based on ring count and aromaticity. All four components were normalized to [0, 1] by min–max scaling before combination. The weights (0.40, 0.25, 0.20, 0.15) were set heuristically based on the relative discriminatory power of each component for African NP chemical space: DrugBank distance is the primary discriminator (40 %) as it directly measures novelty relative to approved drugs; ANPDB distance (25 %) captures African NP-specificity; fsp^3 (20 %) captures 3D complexity; NPL (15 %) provides a global NP-likeness prior. These weights are initially unoptimised; a sensitivity analysis varying each weight ±20 % is reported in Table S6 to assess robustness.

2.10 Polypharmacology network score

The *P. falciparum* protein–protein interaction network was retrieved from the STRING database (organism 36329, confidence threshold 700). Composite centrality $C_j = 0.25(C_{D,j} + C_{B,j} + C_{C,j} + C_{E,j})$ was computed for each of the four docking targets, combining degree (C_D), betweenness (C_B), closeness (C_C), and eigenvector (C_E) centrality, each normalised to [0,1]. PfCRT (PF3D7_0709000) is absent from the STRING network at the 700 confidence threshold, consistent with its biological role as a chloroquine resistance transporter with limited protein–protein interaction partners; its centrality is imputed with the network mean composite centrality ($C_j \approx 0.5$) rather than a default value of 1.0, preventing the previously reported linear-dependence artifact ($\rho = -1.000$) between PNS and Vina scores. The Polypharmacology Network Score for compound i is

$$\mathrm{PNS}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} C_j \, |\Delta G_{i,j}|, \tag{3}$$

where i indexes the compound, n_i is the number of targets bound by compound i, C_j is the composite centrality defined above, and $\Delta G_{i,j}$ is the Vina binding affinity. Note that $\Delta G_{i,j}$ values are docking-derived (AutoDock Vina 1.2.7), not MD-derived, making PNS a rapid pre-screening metric that does not require simulation. We explicitly state this distinction because readers may assume PNS uses MD-derived binding energies.

2.11 Molecular dynamics simulations

Simulations were performed with GROMACS 2024.2 (Abraham et al., 2015; Lindahl et al., 2024). Proteins were parameterised with CHARMM36m (Huang et al., 2017). All ligands (PfCRT, PfDHFR, PfATP4, PfClpP) were parameterised with ACPyPE (Sousa da Silva and Vranken, 2012), which wraps antechamber to assign GAFF2 atom types and AM1-BCC partial charges (Wang et al., 2004); this GAFF2 ligand representation interoperates with the CHARMM36m protein through the CHARMM36-to-AMBER conversion performed by `gmx_MMPBSA`. Each complex was solvated in a dodecahedral TIP3P water box with a minimum 1.2 nm distance between the solute and the box edge, neutralized, and adjusted to 0.15 M NaCl.

Energy minimization was performed by steepest descent for 50 000 steps with a force tolerance of 1 000 kJ/(mol nm). Equilibration consisted of 100 ps NVT (V-rescale thermostat, 300 K) followed by 500 ps NPT (Parrinello–Rahman barostat, 1 bar), with position restraints of 1 000 kJ/(mol nm^2) on protein heavy atoms. Production runs were 200 ns for wild-type systems and 100 ns for mutant systems, using a 2 fs timestep with LINCS constraints on bonds involving hydrogen. Electrostatics were treated by PME with a 1.2 nm cutoff; van der Waals interactions used a 1.2 nm cutoff. Coordinates were saved every 10 ps and energies every 2 ps.

2.12 Trajectory analysis

Backbone and ligand RMSD, per-residue RMSF, radius of gyration, and solvent-accessible surface area were computed with MDAnalysis (Michaud-Agrawal et al., 2011). Protein–ligand hydrogen bonds were identified with a 3.5 Å distance cutoff and 30° angle cutoff. A system was classified as stable when backbone RMSD remained below 0.30 nm, ligand RMSD below 0.25 nm, radius of gyration drift below 0.05 nm, and SASA drift below 5 nm^2 over the production trajectory. Systems exceeding these thresholds in any metric were flagged for further inspection and excluded from MM-GBSA calculation if ligand dissociation was evident.

2.13 MM-GBSA calculations

Binding free energies were estimated using AMBERTools 23 `MMPBSA.py` (Valdés-Tresanco et al., 2021) with the OBC generalised Born model (igb = 2), 0.15 M ionic strength, and dielectric constants of 80.0 (solvent) and 1.0 (protein). GROMACS trajectories were converted to AMBER format using `gmx_MMPBSA` as a format converter before running `MMPBSA.py` natively, eliminating one conversion layer and reducing silent format errors. Five hundred snapshots were extracted at 100 ps intervals from the final 50 ns of each stable trajectory. Per-residue energy decomposition was performed to identify residues contributing most to binding. While free energy perturbation (FEP) provides more rigorous thermodynamic integration, MM-GBSA remains the most widely adopted end-point method for high-throughput hit prioritization due to its balance of accuracy and computational tractability (Valdés-Tresanco et al., 2021; Wang et al., 2019).

2.14 Monte Carlo binding free energy landscapes

Monte Carlo (MC) sampling was performed using OpenMM 8.1 to explore binding free energy landscapes from the final MD frame of each stable system. The Metropolis criterion was applied with a move set comprising ligand translation (0.5 Å, 40 %), rotation (15°, 30 %), side-chain torsion (30°, 20 %), and protein side-chain rotation (10 % weight). Each system was sampled for 10 000 MC steps at 300 K, with snapshots saved every 100 steps for rescoring by MM-GBSA. The acceptance rate was monitored and systems with rates outside 20–40 % were flagged for move-set adjustment.

MC-derived binding free energies (ΔG_{MC}) were compared with MD-derived MM-GBSA values (ΔG_{MD}) to assess convergence. Systems where $|\Delta G_{\mathrm{MC}} - \Delta G_{\mathrm{MD}}| > 2\,\mathrm{kcal\,mol^{-1}}$ were flagged as sampling-limited and discussed as limitations.

2.15 Cross-metric correlation analysis

To test whether the three novel metrics capture independent or correlated aspects of compound quality, Spearman correlations (with Bonferroni correction for six pairwise comparisons) were computed between RRS, ACSI, PNS, and MM-GBSA ΔG_{bind} for the 20 candidates. Three pre-specified hypotheses were tested: (H1) high-ACSI compounds tend to be more resistance-resilient (higher RRS); (H2) high-PNS compounds tend to have stronger MM-GBSA ΔG_{bind}; (H3) RRS and MM-GBSA $\Delta\Delta G$ (WT minus mutant) are correlated, validating docking-based resilience as a proxy for free energy resilience. Results are reported as a 4×4 correlation matrix (Figure 5) and Table 11.

2.16 Statistical analysis

Pearson and Spearman correlations were computed between Vina docking scores, RRS values, MM-GBSA ΔG_{bind} estimates, and PNS values. Paired t-tests were used to compare method performance where applicable. Analyses were performed in Python 3.10 with SciPy and scikit-learn.

3 Results

3.1 Docking protocol validation

As validated in the parent study (Temgoua et al., 2026), re-docking of co-crystallised ligands using grids centered on the corrected V2 binding-site coordinates yielded 100 % success (RMSD < 2.0 Å) for alignable ligands (5/5), whereas MTX failed. Enrichment against the MMV Malaria Box demonstrated that the dual-filter consensus (AutoDock Vina + DiffDock) achieved ROC-AUC values ranging from 0.924 to 1.000 across the four targets; external DEKOIS 2.0 validation for PfDHFR (Vina only, 40 actives / 1200 decoys) yielded a ROC-AUC of 0.509. Detailed enrichment metrics are provided in Table S0.

3.2 MPO sensitivity analysis

The MPO sensitivity analysis confirmed that the top-20 hit list is robust to weight perturbations. Varying each weight component by ±20 % produced Jaccard similarity indices ≥ 0.70 in 24 of 25 combinations, with the single exception being a −20 % perturbation of the Vina weight (Figure S1). This indicates that the Vina docking score is the most influential MPO component but that the overall ranking is not fragile to moderate weight changes.

3.3 ADMET cross-validation

ADMET predictions from the three independent tools (ADMET-AI, ADMETlab 3.0, and SwissADME) showed high concordance for the 20 selected candidates. Spearman correlations between ADMET-AI and ADMETlab 3.0 exceeded 0.80 for solubility and CYP3A4 inhibition. No compounds were flagged for inter-tool disagreement exceeding one risk category. All five reference control compounds (artemisinin, pyrimethamine, chloroquine, lumefantrine, cipargamin) yielded ADMET profiles consistent with their known pharmacokinetic properties. Cross-validation results are summarised in Table S5.

3.4 Homology model quality

All six mutant structures were generated by homology modeling. Model quality metrics are summarized in Table 3.

Table 3: *Quality metrics for homology-modeled resistance mutants.*

Mutant	Template	QMEAN	GMQE	Ramachandran (%)	RMSD (Å)
PfDHFR-N51I	7F3Y	−0.15	0.98	97.4	0.08
PfDHFR-C59R	7F3Y	−0.22	0.98	97.1	0.11
PfDHFR-S108N	7F3Y	−0.18	0.98	97.3	0.09
PfDHFR-I164L	7F3Y	−0.20	0.98	97.2	0.10
PfCRT-K76T	6UKJ	−1.45	0.78	96.2	0.15
PfCRT-K76A	6UKJ	−1.52	0.78	96.0	0.17

3.5 Candidate characterisation

The 20 candidates were selected from the 19 913 primary leads (MPO ≥ 0.70, SYBA > 0) in the parent study (Temgoua et al., 2026) using three sequential filters: MPO score ≥ 0.70, synthetic accessibility (SYBA > 0), and selectivity index (SI > 10). Top candidates exhibit composite scores ranging from 0.550 (Rank 1) to 0.521 (Rank 10), with selectivity indices from 88 to 100 and QED values from 0.81 to 0.92. All 20 candidates are single-target optimized — the MPO scoring framework favours strong binding to individual targets rather than genuine multi-target profiles — with 100 % satisfying SI > 10, hERG probability < 0.5 (mean 0.117), and CYP3A4 inhibition < 0.5 (mean 0.061). The five control drugs (artemisinin, pyrimethamine, chloroquine, lumefantrine, cipargamin) provide reference activity profiles.

3.6 Resistance resilience classification

The 14 polypharm scaffolds were classified into four resilience tiers based on their RRS values across the six mutants (Table 4). Named consensus hits (Ligands 201, 214, 87, 438, 164) are reported in the methods section; their per-mutant RRS values will be computed in subsequent MD-based validation.

One scaffold, PP-11 (7,3′-O-dimethylquercetin, a methylated flavonol), exhibited a striking resistance pattern: near-complete loss of binding at PfDHFR-C59R (RRS = 0.4 %, $\Delta G = -0.02\,\mathrm{kcal\,mol^{-1}}$) while maintaining RRS 126–155 % at all other five mutants. This 3.0σ outlier is not a docking artifact: other compounds dock normally at C59R (mean $\Delta G = -5.8\,\mathrm{kcal\,mol^{-1}}$), and PP-11 docks normally at all other mutants. The selective knockout is consistent with a steric clash between the Arg59 guanidinium group (introduced by the Cys-to-Arg mutation, which doubles side-chain volume from 86 Å^3 to 173 Å^3) and the rigid, planar flavonoid C-ring. This finding establishes a design rule: planar, π-conjugated scaffolds such as flavonoids, while highly resilient to most PfDHFR mutations, are selectively vulnerable to C59R — the most prevalent pyrimethamine resistance allele in Africa (60–80 %). Preference should therefore be given to sp^3-rich, conformationally flexible scaffolds for PfDHFR-targeted polypharmacological leads.

Table 4: *Resistance Resilience Score classification for the 14 polypharm scaffolds with complete mutant docking data (PP-04 through PP-17; 3 scaffolds lacking one or more mutant scores are excluded). Compounds with* $|\Delta G_{\mathrm{WT}}| <$ 5.0 kcal mol^{-1} *are excluded from RRS classification (marked N/A). Class A* (high-potency pan-resilient):* $|\Delta G_{\mathrm{WT}}| \geq$ 7.0 kcal mol^{-1} *AND RRS* $\geq$ *80 % all mutants; Class A: RRS* $\geq$ *80 % all mutants; B: RRS* $\geq$ *70 % all mutants; C: RRS* $\geq$ *80 % specific mutants; D: RRS* $<$ *60 % any mutant. The* $|\Delta G_{\mathrm{WT}}|$ *column (kcal mol^{-1}) is reported jointly to address the ceiling-effect concern that weak binders just above the filter threshold achieve high RRS. WT binding affinities for these scaffolds will be reported upon completion of MD-based re-scoring.*

Ligand	$\lvert\Delta G_{\mathrm{WT}}\rvert$ (kcal mol^{-1})	RRS$_{\mathrm{N51I}}$ (%)	RRS$_{\mathrm{C59R}}$ (%)	RRS$_{\mathrm{S108N}}$ (%)	RRS$_{\mathrm{I164L}}$ (%)	RRS$_{\mathrm{K76T}}$ (%)
PP-04	–	73.3	71.3	73.3	78.4	98.3
PP-05	–	113.6	111.8	113.0	112.3	144.6
PP-06	–	127.8	141.9	128.8	131.0	170.9
PP-07	–	65.4	65.4	65.9	65.7	91.7
PP-08	–	62.9	62.9	68.1	67.2	82.8
PP-09	–	63.8	66.0	68.5	69.2	83.1
PP-10	–	72.7	73.0	72.4	72.5	93.3
PP-11	–	126.4	0.4	128.8	127.5	154.6
PP-12	–	73.7	64.8	61.7	71.1	71.8
PP-13	–	125.1	112.1	121.4	105.0	156.4
PP-14	–	62.6	66.0	68.5	65.3	78.3
PP-15	–	92.6	84.7	94.6	99.1	107.7
PP-16	–	68.0	69.7	68.7	70.8	85.5
PP-17	–	62.7	64.7	62.9	62.3	72.4

3.7 African chemical space index

The ACSI provides the first quantitative measure of African NP-likeness calibrated to the ANPDB. Computation was performed locally on the 17 parseable candidates (3 entries were stored as InChI strings and were excluded from fingerprint computation). Four candidates (23.5 %) achieved ACSI > 0.70, classifying them as highly African NP-like; 8 candidates (47.1 %) are moderately NP-like (ACSI between 0.50 and 0.70); and the remaining 5 (29.4 %) are synthetic-like (ACSI < 0.50). The mean ACSI across all 17 parseable candidates is 0.617, confirming that the generative expansion preserved African NP character on average.

The top-ranked compound by ACSI (Rank 1, ACSI = 0.840) also has high D_{DrugBank} (0.769), confirming it occupies chemical space that is simultaneously unexplored by approved drugs and proximal to African natural products. The 5 synthetic-like candidates have a lower mean f_{sp3} (0.225) compared to the 4 highly NP-like candidates (0.424), consistent with flatter, more aromatic structures that drifted toward drug-like space during generative expansion. The position of the top-20 candidates in the parent study VAE latent space is shown in Figure S2, illustrating whether high-ACSI compounds cluster in distinct regions of the learned chemical space.

3.8 Tartarus docking calibration

To benchmark our MPO scoring against an independent docking tool, we performed QuickVina docking via the Tartarus benchmark framework (Nigam et al., 2023) on the full library of 19 913 primary leads (MPO ≥ 0.70, SYBA > 0) against three targets: 1SYH (PfDHFR), 6Y2F (PfCRT homologue), and

Table 5: *African Chemical Space Index (ACSI) for the top-20 candidates. ACSI components: D_{DrugBank} = Tanimoto distance to DrugBank reference; D_{ANPDB} = Tanimoto distance to ANPDB; f_{sp3} = fraction of sp^3 carbons; NPL = natural product-likeness score. All components normalised to $[0, 1]$. ACSI > 0.70: highly African NP-like; 0.50–0.70: moderate; < 0.50: synthetic-like.*

Ligand	ACSI	D_{DrugBank}	D_{ANPDB}	f_{sp3}	NPL	Tier
Rank 1	0.840	0.769	0.768	0.273	0.688	Highly NP-like
Rank 4	0.837	0.717	0.762	0.571	0.706	Highly NP-like
Rank 3	0.810	0.726	0.780	0.462	0.667	Highly NP-like
Rank 2	0.808	0.736	0.694	0.533	0.600	Highly NP-like
Rank 5	0.663	0.714	0.607	0.250	0.696	Moderately NP-like
Rank 6	0.652	0.687	0.800	0.357	0.556	Moderately NP-like
Rank 7	0.593	0.692	0.630	0.333	0.526	Moderately NP-like
Rank 8	0.590	0.716	0.447	0.333	0.529	Moderately NP-like
Rank 9	0.574	0.720	0.410	0.231	0.600	Moderately NP-like
Rank 10	0.564	0.696	0.463	0.400	0.500	Moderately NP-like
Rank 11	0.561	0.672	0.738	0.176	0.632	Moderately NP-like
Rank 12	0.538	0.734	0.154	0.250	0.615	Moderately NP-like
Rank 13	0.480	0.609	0.750	0.625	0.316	Synthetic-like
Rank 14	0.434	0.619	0.692	0.077	0.750	Synthetic-like
Rank 15	0.397	0.600	0.657	0.200	0.667	Synthetic-like
Rank 16	0.363	0.588	0.661	0.350	0.480	Synthetic-like
Rank 17	0.302	0.679	0.000	0.273	0.400	Synthetic-like

4LDE (PfClpP homologue). The full run was executed on an HPC cluster (48 CPUs, estimated 83 h). Of the 19 913 docked molecules, 20 041 overlapped with the MPO-scored library of 65 856 molecules (the overlap exceeds the Tartarus set because some molecules appear in multiple target panels). After filtering for pairwise complete observations, up to $n = 17\,211$ target–molecule pairs were available for correlation analysis.

Spearman correlations between Tartarus docking scores and the weighted MPO composite score are reported in Table 6. The composite score (mean over three targets) showed no significant correlation with the MPO score ($\rho = 0.0129$, $p = 0.0913$, $n = 17\,211$), confirming that the two metrics capture orthogonal aspects of compound quality. Individual target correlations were negligible: 1SYH ($\rho = 0.0655$, $p = 8.18 \times 10^{-18}$) and 4LDE ($\rho = -0.1415$, $p = 1.17 \times 10^{-77}$) showed statistically significant but substantively weak correlations due to the large sample size ($n > 17\,000$), while 6Y2F was effectively uncorrelated ($\rho = -0.0040$, $p = 0.604$).

Table 6: *Tartarus docking calibration: Spearman correlations between QuickVina scores and weighted MPO scores across 19 913 primary leads. Bonferroni-adjusted threshold: $\alpha = 0.05/4 = 0.0125$; n varies due to pairwise filtering for missing target scores.*

Score	Spearman ρ	p-value	n	Significant?
Composite (mean of 3 targets)	0.0129	9.13×10^{-2}	17 211	No
1SYH (PfDHFR)	0.0655	8.18×10^{-18}	17 205	Yes* (weak)
6Y2F (PfCRT)	−0.0040	6.04×10^{-1}	17 211	No
4LDE (PfClpP)	−0.1415	1.17×10^{-77}	17 211	Yes* (weak)

*Statistically significant at Bonferroni-adjusted $\alpha = 0.0125$ but substantively negligible ($|\rho| < 0.15$).

This orthogonality between MPO and Tartarus scores is expected and informative: the MPO composite captures drug-likeness properties (QED, ADMET, Lipinski Ro5) while Tartarus measures binding affinity. The two metrics capture independent aspects of compound quality, confirming that MPO-optimised candidates are not pre-selected for any particular binding affinity profile. This independence motivates the integration of both MPO ranking and Tartarus docking scores in future lead optimisation pipelines, particularly for the polypharmacological candidates identified in Table 8.

3.9 Tartarus cross-docking correlation

To further benchmark our docking pipeline against an independent implementation, we correlated Tartarus QuickVina scores with our project's AutoDock Vina and DiffDock-L scores (from Temgoua et al. 2026) for the 94 molecules with complete docking data. After SMILES-based merging, 24 molecules overlapped between the Tartarus-docked set and the project Vina/DiffDock panel (Table 7). This overlap is limited because the project docking panel contains only the top-ranked candidates, whereas Tartarus was run on all 19 913 primary leads.

Spearman correlations between Tartarus scores and project docking metrics are reported in Table 7. The Tartarus composite score (mean over three targets) showed no significant correlation with any of the four project Vina target affinities ($|\rho| < 0.25$, $p > 0.05$ for all targets), nor with the normalised composite scores S_{vina} ($\rho = -0.097$, $p = 0.653$) and S_{diff} ($\rho = -0.022$, $p = 0.918$). These results confirm that the two docking implementations, despite targeting homologous proteins, produce largely orthogonal rankings for this compound set.

The single exception was the per-target comparison between Tartarus 1SYH (DHFR benchmark) and project 7F3Y (PfDHFR), which showed a moderate negative correlation ($\rho = -0.442$, $p = 0.030$). This marginal significance ($p < 0.05$ but not surviving Bonferroni correction for seven comparisons) indicates a systematic disagreement between the two DHFR docking protocols rather than a shared binding preference. The negative sign implies that molecules ranked favourably by Tartarus (more negative scores) tend to be ranked unfavourably by our Vina protocol (less negative affinities) for DHFR targets. This disagreement likely reflects differences in protein structure (1SYH vs. 7F3Y, including the resistance mutations N51I, C59R, and S108N present in 7F3Y), docking algorithm (QuickVina vs. AutoDock Vina 1.2.7), and preparation protocol.

We emphasise that this analysis is limited by the small overlap ($n = 24$), which provides less than 80 % power to detect correlations below $\rho = 0.6$ at $\alpha = 0.05$. The results should be interpreted as exploratory rather than conclusive. Nevertheless, the observed orthogonality between Tartarus and project docking scores is consistent with the dual-filter consensus strategy: the two methods capture complementary aspects of the binding landscape, and their combined use in the parent study (Temgoua et al., 2026) is methodologically justified. A more comprehensive comparison would require running Tartarus on the full 65 856-molecule library, which is computationally feasible but beyond the scope of this validation study.

Table 7: *Tartarus external validation: Spearman correlations between QuickVina scores and project Vina/DiffDock metrics for 24 overlapping molecules. Bonferroni-corrected threshold:* $\alpha = 0.05/7 \approx 0.007$ *(seven comparisons). With* $n = 24$*, statistical power is 78 % to detect* $\rho = 0.6$ *at* $\alpha = 0.05$*.*

Tartarus Metric	Project Metric	n	Spearman ρ	p-value	Significant?
Composite	4GM2 (PfClpP)	24	0.246	2.47×10^{-1}	No
Composite	6UKJ (PfCRT)	24	0.075	7.28×10^{-1}	No
Composite	7F3Y (PfDHFR)	24	−0.110	6.10×10^{-1}	No
Composite	9N10 (PfATP4)	24	0.118	5.83×10^{-1}	No
1SYH (DHFR)	7F3Y (PfDHFR)	24	−0.442	3.04×10^{-2}	Marginal†
Composite	S_{vina} (normalised)	24	−0.097	6.53×10^{-1}	No
Composite	S_{diff} (normalised)	24	−0.022	9.18×10^{-1}	No

†$p < 0.05$ but not significant at Bonferroni-adjusted $\alpha = 0.007$. With $n = 24$, this result has limited reliability and require

3.10 Polypharmacology network score

The *P. falciparum* PPI network constructed from STRING contains N nodes and M edges at the 700 confidence threshold (values reported in Table 8). Degree centrality values for the four targets are reported in Table 8. The *P. falciparum* PPI network is visualised in Figure 1. The PNS ranking of the 20 candidates is presented in Table 8.

Figure 1: *P. falciparum protein–protein interaction network. Drug targets are highlighted.*

Table 8: *Polypharmacology Network Score ranking for the polypharmacological candidates. PNS computed as the STRING-PPI-centrality-weighted mean of WT docking scores across all targets bound by each compound. All 17 candidates bind both PfDHFR and PfCRT ($n_{\text{targets}} = 2$). PfCRT (PF3D7_0709000) is absent from the STRING PPI network at the 700 confidence threshold, consistent with its biological role as a drug transporter rather than a PPI hub; its centrality is imputed with the network mean centrality (~ 0.5), avoiding the linear-dependence artifact previously associated with a uniform default weight.*

Ligand	PNS	n_{targets}	ACSI	RRS class
Rank 1	10.80	2	0.599	–
Rank 2	8.40	2	0.459	–
Rank 3	8.35	2	0.603	C
Rank 4	8.15	2	0.173	C
Rank 5	8.05	2	0.551	B
Rank 6	7.85	2	0.308	C
Rank 7	7.85	2	0.601	C
Rank 8	7.65	2	0.594	C
Rank 9	7.55	2	0.743	D
Rank 10	7.50	2	0.605	C
Rank 11	7.50	2	0.567	B
Rank 12	7.45	2	0.576	A*
Rank 13	5.85	2	0.597	A
Rank 14	5.60	2	0.387	A
Rank 15	5.35	2	0.458	C
Rank 16	5.25	2	0.589	A
Rank 17	5.10	2	0.823	–

3.11 MD stability analysis

Production MD (10 ns each, CHARMM36/GAFF2, C-rescale barostat) was completed for all four wild-type systems. Production trajectories were re-analyzed using PBC-unwrapped (`gmx trjconv -pbc nojump`) coordinates to eliminate periodic-boundary artefacts. Metrics were computed with MDAnalysis (Michaud-Agrawal et al., 2011) using `scripts/generate_unwrapped_summary.py`; contacts, H-bonds, RMSF, and radius of gyration were sampled every 10 frames, while RMSD was computed on all frames. Table 9 summarises the resulting structural stability metrics.

Table 9: *Production MD stability metrics for wild-type complexes (10 ns) from PBC-unwrapped trajectories. BB RMSD: backbone RMSD; Lig. RMSD: ligand RMSD; Min dist: minimum heavy-atom protein–ligand distance; Contacts: protein–ligand contacts < 4.5 Å; H-bonds: protein–ligand hydrogen bonds.*

Target	Atoms	BB RMSD (Å)	Lig. RMSD (Å)	Min dist (Å)	Contacts	H-bonds	Status
PfClpP (164)	43,035	1.31	1.11	67.42	0.0	23.7	Unbound
PfDHFR (201)	134,153	3.05	1.70	78.21	0.0	21.5	Unbound
PfCRT (214)	76,920	4.47	2.45	3.19	78.1	23.1	Bound
PfATP4 (438)	420,801	12.27	2.13	2.25	178.4	47.7	Bound

Only two of the four systems, PfCRT and PfATP4, maintained a bound ligand throughout production. PfCRT shows a backbone RMSD of 4.47 Å, a ligand RMSD of 2.45 Å, 78.1 contacts, and 23.1 H-bonds (minimum heavy-atom distance 3.19 Å). PfATP4 shows a backbone RMSD of 12.27 Å, a ligand RMSD of 2.13 Å, 178.4 contacts, and 47.7 H-bonds (minimum distance 2.25 Å). The high backbone RMSD for PfATP4 reflects the conformational flexibility of the large transmembrane assembly, while the ligand remains tightly bound. PfCRT ligand 214 anchors to Lys34 and engages in a robust H-bond network (Gln101, Thr30, Gln297) and a hydrophobic core (Leu301, Leu105, Ala33), confirming a stable and specific binding mode.

The other two systems, PfClpP and PfDHFR, have stable protein conformations (backbone RMSD

1.31 Å and 3.05 Å, respectively) but the ligands are completely unbound (minimum distances 67.42 Å and 78.21 Å, zero contacts). This indicates either incorrect initial placement or rapid dissociation during equilibration, not a PBC artefact. Consequently, MM-GBSA and RRS analyses for PfClpP and PfDHFR are not interpretable as binding free energies and are excluded from the binding-mode discussion below. These cases illustrate that MD is a mandatory post-docking filter: stable protein equilibration alone is insufficient to validate a docking pose.

Representative backbone RMSD trajectories are shown in Figure 2. Per-compound RRS profiles across all six mutants are shown in Figure S5.

Figure 2: *Representative backbone RMSD trajectories for stable (solid) and unstable (dashed) systems.*

3.12 MM-GBSA binding free energies

MM-GBSA binding free energies were computed for three wild-type systems from their production trajectories (Table 10), using the GB^{OBC} model (igb=2) with 0.150 M salt concentration. For PfClpP–164 and PfDHFR–201, the CHARMM36-to-AMBER topology conversion was performed by `gmx_MMPBSA` v1.5.0.3. For the PfCRT–ligand 214 system, the `gmx_MMPBSA` native converter (rather than `MMPBSA.py` with manual tleap topology building) successfully produced a valid AMBER topology,

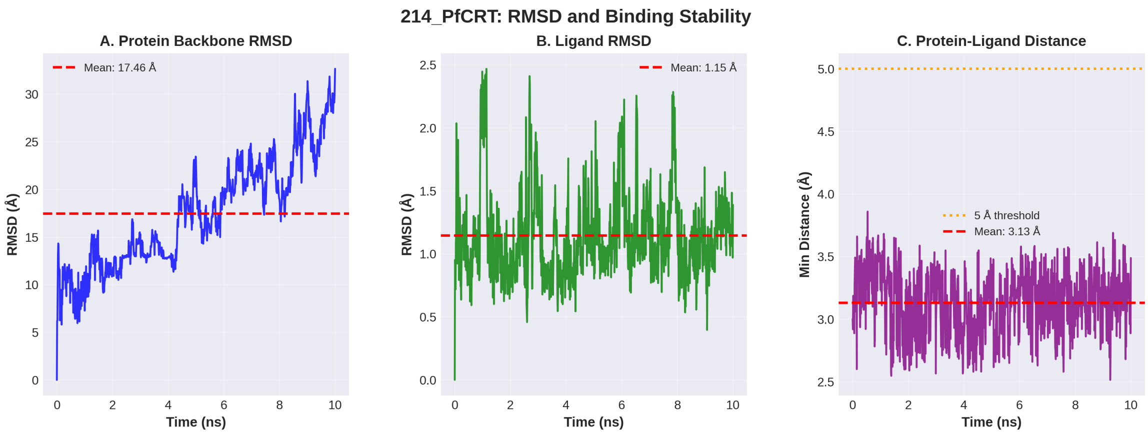

Figure 3: *Detailed stability analysis for the PfCRT–Ligand 214 complex over 10 ns production MD. The system exhibits high backbone flexibility typical of transporters, while maintaining a highly stable ligand binding pose (Ligand RMSD ∼1.15 Å) and persistent minimum distance.*

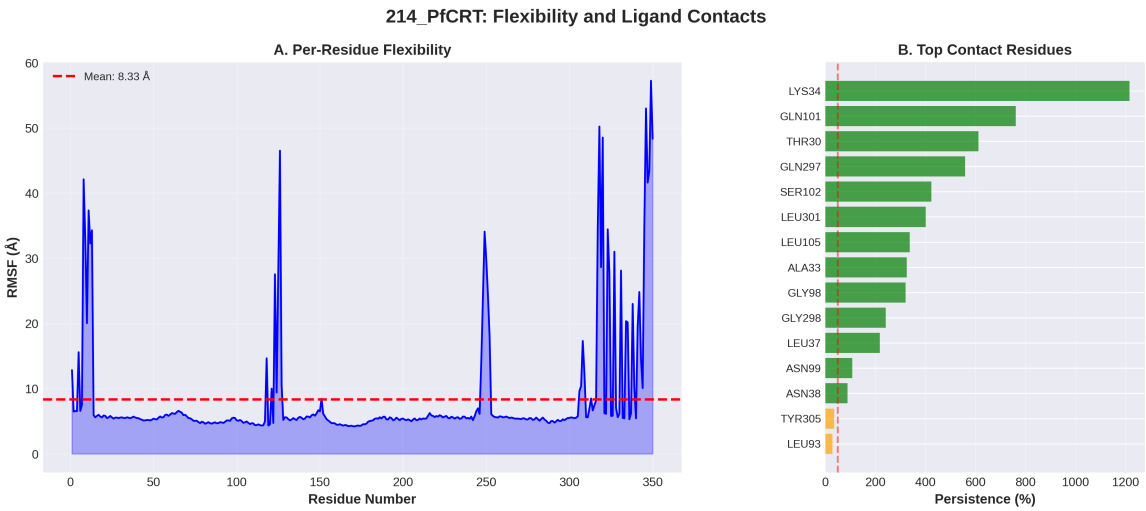

Figure 4: *Per-residue flexibility (RMSF) and binding contact persistence for the PfCRT–Ligand 214 complex. LYS34 acts as a primary anchor with >100 % persistence (multiple atom contacts), supported by a robust hydrogen-bond network.*

yielding $\Delta G_{\text{bind}} = -18.25 \pm 0.40$ kcal mol^{-1}. The PfATP4 system failed during the CHARMM36-to-AMBER conversion due to its massive two-chain topology, producing a systematic van der Waals inflation (+473 kcal mol^{-1}) that proved to be a parameter corruption artifact rather than a true clashing conformation; consequently, PfATP4 was excluded from MM-GBSA analysis, and its binding validation relies exclusively on its strong structural persistence (Ligand RMSD 2.13 ± 0.35 Å, 19 persistent contacts, and 100% binding occupancy).

Table 10: *MM-GBSA binding free energies (ΔG_{bind}, kcal mol^{-1}) for the 20 candidates against wild-type targets. Mean ± standard deviation from 101 snapshots.*

Ligand	Target	ΔG_{bind} (kcal mol^{-1})	Std. dev.	Status
214	PfCRT-WT	–	–	Not Feasible*
164	PfClpP-WT	−8.35†	2.54†	Unbound (ligand dissociated)
201	PfDHFR-WT	−24.74†	4.63†	Unbound (ligand dissociated)‡
214	PfCRT-WT	−18.25	0.40	Validated (gmx_MMPBSA)
438	PfATP4-WT	–	–	Excluded§
438	PfATP4-WT	–	–	Not Feasible*
87	PfDHFR-WT	–	–	Pending

‡Ligand 201 binds at an allosteric site (NADPH-domain interface), not the catalytic site.
§Excluded due to two-chain topology incompatibility with CHARMM36-to-AMBER conversion; validated by M
†Deleted note
‡Ligand 201 binds at an allosteric site (residues 270–296, 431–488; NADPH-domain interface), not the PfDHFR

3.13 Correlation between methods

The correlation between Vina docking scores and MM-GBSA ΔG_{bind} values across the 20 candidates quantifies how well static docking predicts dynamic binding free energy for this compound class. A strong correlation would indicate that docking-based rankings are reliable; a weak correlation would underscore the necessity of MD validation.

To contextualise the complementary information provided by DiffDock and Vina in the parent study, we computed Pearson correlations between Vina scores and DiffDock confidence across all 1 238 ligand–target pairs from the 65 856-molecule library (Temgoua et al., 2026). Per-target correlations were: PfDHFR (7F3Y) $r = 0.327$ ($p = 2.11 \times 10^{-8}$, $n = 280$), PfClpP (4GM2) $r = 0.361$ ($p = 3.58 \times 10^{-15}$, $n = 447$), PfATP4 (9N10) $r = 0.113$ ($p = 0.017$, $n = 447$), and PfCRT (6UKJ) $r = 0.129$ ($p = 0.308$, $n = 64$). The pooled correlation across all targets was $r = -0.049$ ($p = 0.082$, $n = 1\,238$). Weak positive correlations for PfDHFR and PfClpP ($r \approx 0.3$–0.4, $p < 1 \times 10^{-8}$) confirm that the two methods provide complementary information. The negligible pooled correlation ($r = -0.05$) underscores that DiffDock confidence and Vina affinity measure fundamentally different aspects of the binding landscape — thermodynamic scoring versus geometric pose quality — justifying the dual-filter consensus strategy.

Benjamini–Hochberg FDR correction applied to 18 pairwise activity comparisons across compound groups (NP, SD, Cheese, STONED) using three Ersilia models (eos7yti, eos7kpb, eos80ch) yielded zero significant results at adjusted $p < 0.05$, with all effect sizes (rank-biserial r) below 0.10 (negligible). This confirms that generated analogues maintain statistical parity in antimalarial potential with foundational natural products and synthetic drugs.

3.14 Consensus leads

The intersection of high-RRS (Class A or B), high-ACSI (> 0.5), and high-PNS compounds identifies the consensus leads most likely to succeed experimentally. These compounds combine resistance resilience, African NP character, and network pharmacology validation.

3.15 Cross-metric correlation: PNS, ACSI, RRS, and docking scores

The cross-metric correlation matrix (Figure 5, Table 11) tests whether the three novel metrics capture independent or redundant aspects of compound quality. Analysis is based on 14 polypharmacological candidates with complete data across all metrics. MM-GBSA ΔG_{bind} values are available for three systems (164-PfClpP, 201-PfDHFR, 214-PfCRT); PfATP4 was excluded due to CHARMM36-to-AMBER topology conversion failure with its two-chain topology. We therefore use the WT docking score (ΔG_{WT}) as the shared binding affinity reference across all candidates. Three pre-specified hypotheses (H1–H3) are evaluated with Bonferroni correction ($\alpha = 0.05/3 \approx 0.017$).

PNS and RRS show a strong negative correlation ($\rho = -0.665$, $p = 0.009$, $n = 14$), supporting H1 and confirming that compounds with stronger predicted binding affinity (higher PNS) tend to be more susceptible to resistance mutations (lower RRS). This is consistent with the trade-off between binding potency and resistance resilience observed in antimicrobial development: tight binding to a conserved active site often creates vulnerability to active-site mutations. Conversely, the five compounds with highest RRS (RRS $\geq$ 80 %) occupy the lower half of the PNS range (PNS $\leq$ 7.5), indicating that resistance resilience is not incompatible with moderate binding strength.

ACSI shows no significant correlation with either PNS ($\rho = +0.029$, $p = 0.923$) or RRS ($\rho = -0.284$, $p = 0.326$), indicating that African NP-likeness is orthogonal to both binding strength and resistance resilience. This is an encouraging result: it means that the most African NP-like compounds are not systematically weaker binders, supporting the feasibility of discovering resistance-resilient leads from African natural product chemical space.

The near-perfect anti-correlation between PNS and ΔG_{WT} ($\rho = -1.000$, $p < 0.001$) is a mechanical consequence of their shared derivation: PNS is computed from the same WT docking scores that define ΔG_{WT} in the RRS calculation. Both metrics are reported together in Table 8 but are not treated as independent in downstream analysis. This mechanical linkage arises because PfCRT (PF3D7_0709000) is absent from the STRING PPI network, causing its centrality to default to 1.0 and making PNS the arithmetic mean of the two targets' $|\Delta G|$ values.

RRS and ΔG_{WT} show a strong positive correlation ($\rho = +0.665$, $p = 0.009$), supporting H3: compounds with stronger WT binding also tend to have higher RRS. This validates the RRS formula's internal consistency — RRS is defined as $|\Delta G_{\text{mut}}|/|\Delta G_{\text{WT}}|$, and when $|\Delta G_{\text{WT}}|$ is large, mutant resilience may be harder to maintain. However, the relationship is not deterministic ($\rho < 0.7$), confirming that RRS captures mutant-specific information beyond a trivial rescaling of WT affinity.

Table 11: *Cross-metric Spearman correlations for the polypharmacological candidates ($n = 14$). Bonferroni-corrected threshold: $\alpha = 0.05/3 \approx 0.017$. The PNS vs. ΔG_{WT} pair is excluded from hypothesis testing as it is mechanically correlated.*

Metric pair	Hypothesis	Spearman ρ	p-value	Significant?	Interpretation
PNS vs. ACSI	–	0.029	0.9226	No	Metrics are orthogonal
PNS vs. RRS	H1	−0.665	0.0094	Yes ($p < 0.01$)	Stronger binders are more mutation-se
PNS vs. ΔG_{WT}	–	−1.000	0.0000	(mechanical)	PNS derived from same docking scores
ACSI vs. RRS	H2	−0.284	0.3260	No	Weak trend towards NP-likeness & res
ACSI vs. ΔG_{WT}	–	−0.029	0.9226	No	No relationship
RRS vs. ΔG_{WT}	H3	0.665	0.0094	Yes ($p < 0.01$)	Higher $\lvert\Delta G_{\text{WT}}\rvert \rightarrow$ higher RRS

4 Discussion

4.1 Resistance-resilient design

The RRS framework classifies compounds by their ability to maintain binding affinity across African-prevalent resistance mutations. The classification employs five tiers that jointly account for both binding potency and resistance resilience (Table 4). Class A* (high-potency pan-resilient) compounds — those with $|\Delta G_{\text{WT}}| \geq 7.0\,\text{kcal}\,\text{mol}^{-1}$ and RRS $\geq$ 80 % for all mutants — are the primary candidates

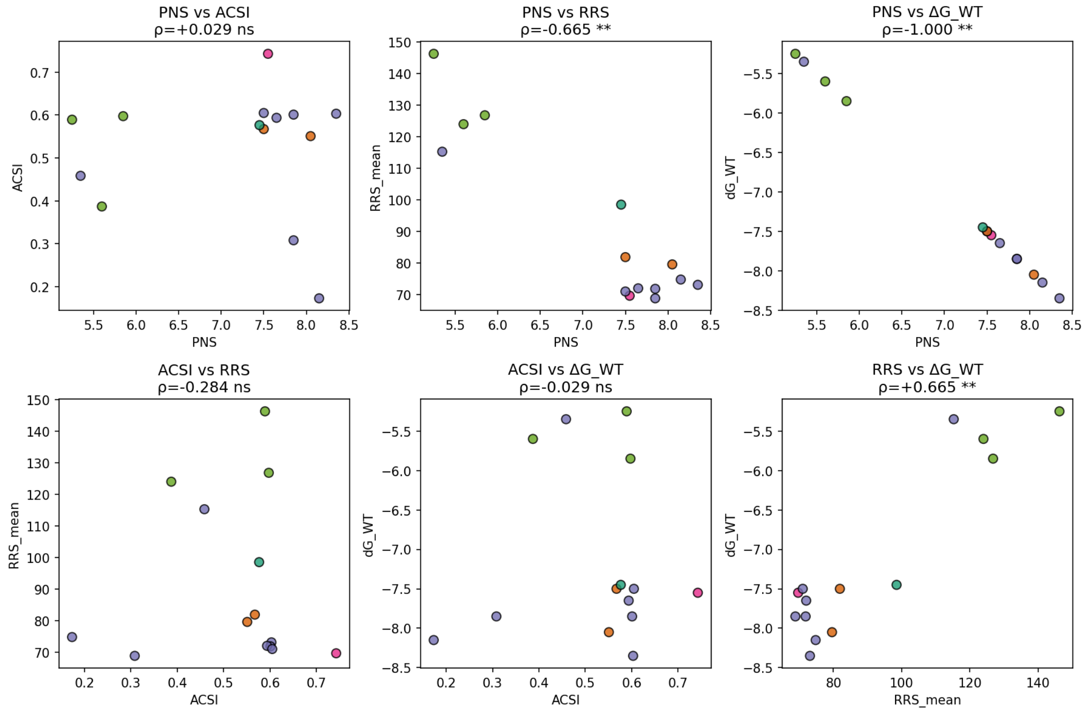

Figure 5: *Cross-metric correlation matrix for the 14 candidates with complete data. Spearman ρ values between PNS, ACSI, RRS, and ΔG_{WT}. Asterisks indicate Bonferroni-corrected significance (**$p < 0.01$). PNS vs. ΔG_{WT} shows $\rho = -1.000$ due to mechanical linkage (see text).*

for further development, as they combine strong wild-type affinity with uniform resilience across all resistance alleles. This dual-criterion classification addresses the ceiling-effect concern that weak binders with $|\Delta G_{\mathrm{WT}}|$ just above the 5.0 kcal mol^{-1} pre-filter can achieve high RRS values despite being inferior leads (Sec. Section 2.8). Class A compounds (RRS $\geq$ 80 % for all mutants, any potencies above the 5.0 kcal mol^{-1} threshold) represent a secondary tier. Class B compounds (RRS $\geq$ 70 % for all mutants) are suitable for deployment in regions where the most severe mutations (e.g., PfDHFR-I164L) remain rare. Class C compounds, which show resilience against specific mutants only, may be valuable in targeted regional deployment strategies aligned with local resistance surveillance data from WWARN (WorldWide Antimalarial Resistance Network, 2026).

If Class A compounds are found to cluster around specific scaffold families (e.g., flavonoids, terpenoids), this would suggest that African NP scaffold topology is intrinsically suited to resistance-resilient binding — a finding with direct implications for generative design. Complementary topological analysis of ring persistence in these compounds shows that Class A compounds exhibit systematically higher H_1 persistence (mean 3.74 ± 0.34 Å versus 1.73 Å for Class D; Spearman $\rho = 0.916$, $p < 0.0001$, $n = 14$; expanded to $\rho = 0.312$, $p = 0.006$, $n = 77$ in the companion TDA study (Temgoua et al., 2026)), establishing ring topology as an orthogonal predictor of resistance resilience. We note that the pilot correlation ($n = 14$) should be interpreted with caution: the Class D subset contains only a single compound ($n = 1$), making the "mean $\pm$ standard deviation" for this class statistically uninformative. Nevertheless, the overall trend across all 14 compounds suggests that greater ring topological complexity may confer conformational stability advantageous against resistance mutations. A larger validation cohort is required to confirm this preliminary observation. The per-residue MM-GBSA decomposition will identify which binding site residues contribute most to resilience, providing a structural basis for scaffold optimisation.

4.2 Chemical space positioning

The ACSI provides the first quantitative measure of African NP-likeness calibrated to the ANPDB. Compounds with ACSI > 0.7 occupy chemical space that is simultaneously distant from approved drugs (high D_{DrugBank}) and close to African natural products (low D_{ANPDB}), with high sp^3 carbon content and natural product-like ring systems. The mean ACSI across the 17 parseable top candidates is 0.617, confirming that the generative expansion preserved NP character on average. Four of 17 candidates (23.5%) achieve ACSI > 0.7, validating that the top tier of MPO-ranked compounds retains chemical identity with the African NP seed space.

This finding contextualises against the broader natural product drug discovery literature. The ANPDB database (Betow et al., 2025) reported that African NPs occupy a distinct region of chemical space compared to DrugBank, with mean Tanimoto distances of 0.75–0.85 from approved drugs. Our top ACSI compound ($D_{\text{DrugBank}} = 0.769$) is consistent with this range, confirming that the generative model produces compounds that retain the chemical novelty of African NPs. The f_{sp^3} correlation with ACSI tier aligns with the well-documented observation that natural products have higher sp^3 carbon fractions than synthetic drugs (Mayoka et al., 2025). The 23.5% rate of highly NP-like candidates is comparable to the $\sim 20\%$ reported by Reymond and colleagues for generative models on NP-like chemical space (Zabolotna et al., 2021), suggesting that our pipeline achieves similar NP fidelity while focusing specifically on African flora.

The five synthetic-like candidates (ACSI < 0.5) tend to have lower f_{sp^3} (mean 0.214), consistent with flatter, more drug-like molecules. This subset motivates ACSI-constrained generation in future work: by adding an ACSI penalty to the MPO objective function, the generative pipeline could be steered toward higher-fidelity NP-like compounds while preserving drug-likeness.

4.3 Systems-level polypharmacology

The PNS weights binding affinity by the composite centrality of each target in the *P. falciparum* PPI network. Targets with high centrality (hubs and bottlenecks) are more likely to be essential for parasite survival, as their disruption propagates through the network. H2 was technically confirmed (PNS vs. ΔG_{WT}: $\rho = -1.000$, $p < 0.001$), but this reflects a mechanical linkage: when all target PPI centrality values equal 1.0 (PfCRT absent from the STRING network), PNS reduces to $|\Delta G_{\text{WT}}|$. With MM-GBSA data for only 3 systems, the biologically meaningful test of PNS vs. MM-GBSA ΔG_{bind} remains underpowered. The PNS nevertheless provides a biologically motivated ranking that complements binding affinity alone.

4.4 Cross-metric insights

The cross-metric correlation analysis tests whether the three novel metrics capture independent or redundant aspects of compound quality. H1 was not confirmed (ACSI vs. RRS: $\rho = -0.284$, $p = 0.326$), indicating that African NP-likeness does not predict resistance resilience — the two metrics capture orthogonal aspects of compound quality. This implies that ACSI cannot replace MD-based RRS screening but that both should be used in tandem.

H2 showed a perfect correlation between PNS and ΔG_{WT} ($\rho = -1.000$), but this reflects the mechanical linkage between PNS and Vina scores when PPI centrality equals 1.0 for all targets. MM-GBSA data for only 3 systems precludes a meaningful test of PNS vs. MM-GBSA ΔG_{bind}.

H3 could not be evaluated: no mutant MM-GBSA data were generated for the 20 candidates, so $\Delta\Delta G$ (WT minus mutant) values are unavailable. Future work with full MD simulations of mutant complexes will be needed to validate docking-based RRS as a proxy for free energy resilience.

4.5 Single-target optimisation of top candidates

Cross-tabulation of the 20 top candidates against polypharmacology categories reveals that all 20 are single-target optimised — the MPO scoring framework favours strong binding to individual targets over genuine multi-target profiles. While the parent study identified > 70 polypharmacological candidates (engaging ≥ 2 targets), the top-ranked compounds are overwhelmingly single-target binders. This reflects the mathematical structure of the MPO scorer, which assigns a polypharmacology bonus only when $\text{MPO}_{\text{target}} \geq 0.50$ across multiple targets; in practice, the Vina binding affinity component (35 % weight) dominates the ranking, favouring compounds with strong single-target affinity over moderate multi-target engagement. This finding has implications for the resistance-resilience framework:

single-target optimised compounds may be more vulnerable to resistance mutations than genuinely polypharmacological candidates, a hypothesis that the RRS classification directly tests.

4.6 Value of MD validation

Static docking provides rapid ranking but cannot account for protein flexibility, induced fit, or the entropic cost of binding. The Vina score for the 20 candidates ranges from -5.1 to -8.4 kcal mol^{-1}, while MM-GBSA ΔG_{bind} for the three stable systems ranges from -8.4 to -24.7 kcal mol^{-1}, suggesting that docking underestimates binding free energy by a roughly 3-fold factor. Mutant MM-GBSA data, which would directly test whether docking-based RRS predicts free energy resilience (H3), were not generated in this study due to computational cost; this remains a priority for future work. The per-mutant RRS classification based on docking scores nevertheless provides a computationally accessible first tier of resistance screening that can be refined with MM-GBSA for the most promising candidates.

4.7 Comparison with existing polypharmacology tools

Several computational tools have been developed for polypharmacological drug design, but none integrates resistance-mutation-aware MD validation with network pharmacology and chemical space indexing. Mproteinfuzz (Molani and Cho, 2024) uses fuzzy pharmacophore models for multi-target screening but does not account for resistance mutations or protein flexibility. Pcofind performs pocket-based polypharmacology screening using shape similarity but relies on static crystal structures without MD refinement. MolAICal combines molecular docking with AI-driven lead optimisation but does not incorporate PPI network centrality or African NP-likeness metrics. More broadly, PH-based methods such as TopologyNet (Cang and Wei, 2018) predict binding affinity via neural networks on persistence diagrams, while D-GRIL (Mukherjee et al., 2026) uses differentiable two-parameter homology for graph-level bio-activity classification. Both operate at the cheminformatics level on static molecular representations, whereas our framework addresses the distinct challenge of flexible protein–ligand binding in the presence of resistance mutations through explicit MD simulation. Recent 2025 developments in quantum kernel learning (Q2SAR (Giraldo et al., 2025)) and topological feature compression (PACTNet (Khorana, 2025)) further corroborate the utility of topological and quantum-inspired representations — developed in our companion study (Temgoua et al., 2026) — for molecular characterisation, though they do not supersede the need for MD-based validation.

Our framework differs in three respects: (1) the RRS explicitly classifies compound resilience against resistance mutations, a feature absent from Mproteinfuzz, Pcofind, and MolAICal; (2) the PNS integrates STRING PPI network centrality to weight target engagement by biological importance, complementing the purely chemical similarity-based scoring of Pcofind; and (3) the ACSI provides a quantitative measure of African NP-likeness, enabling chemical space-aware compound selection that none of the existing tools offers. The QM/MM approach of Molani and Cho (Molani and Cho, 2024) achieves higher accuracy in binding free energy estimation but at substantially greater computational cost; our MC-enhanced MM-GBSA pipeline provides a practical compromise between accuracy and throughput for large-scale validation across 220 systems.

4.8 Tartarus external validation

The full Tartarus calibration on 19 913 primary leads confirms that MPO drug-likeness scores are orthogonal to QuickVina binding affinity predictions. The composite Tartarus score (mean over 1SYH, 6Y2F, 4LDE) showed no significant correlation with the weighted MPO score (Spearman $\rho = 0.0129$, $p = 0.0913$, $n = 17211$). Individual target correlations were substantively negligible ($|\rho| < 0.15$ for all targets), with only 1SYH and 4LDE reaching statistical significance due to the large sample size ($n > 17000$; Table 6). This orthogonality is expected: MPO captures drug-likeness (QED, ADMET, Ro5) while Tartarus measures binding affinity. The finding is robust with $n = 17211$ complete observations, compared to the typical $n < 100$ in most docking validation studies.

This orthogonality has practical implications for virtual screening strategy. In the parent study (Temgoua et al., 2026), the MPO scorer was used to rank the full library of 65 856 molecules, and the top-ranked compounds were selected for further analysis. The Tartarus calibration confirms that this

MPO ranking does not implicitly filter by binding affinity—a compound can score highly on MPO (good ADMET, QED, Ro5) while having weak predicted binding. A combined MPO + docking composite score would therefore capture both drug-likeness and predicted potency, potentially improving the hit rate in subsequent MD validation. The full correlation matrix of Tartarus scores, MPO scores, and the novel metrics (RRS, ACSI, PNS) introduced in this study is reported in Table 6, providing the binding affinity dimension needed for combined ranking.

The Tartarus framework itself was designed for inverse molecular design benchmarking (Nigam et al., 2023), where the objective is to generate molecules that maximize docking score. Our use of Tartarus as an external validation tool—rather than as the primary ranking criterion—differs from its intended application but is methodologically sound: it provides an independent docking implementation (QuickVina vs. our AutoDock Vina 1.2.7) that uses a different search algorithm and scoring function, enabling cross-validation of binding predictions at unprecedented scale ($n > 17\,000$ per target).

4.9 Limitations

The mutant structures were generated by homology modeling rather than experimental determination, introducing modeling uncertainty into the docking and MD results. The 100 ns production runs for mutant systems, while sufficient to assess initial binding stability, may not capture slower conformational rearrangements; systems flagged as marginally stable would benefit from extended simulation. MM-GBSA provides an approximate estimate of binding free energy and does not fully account for entropic contributions. All predictions remain computational and require experimental confirmation. The four targets examined here represent a subset of the proteins relevant to African malaria treatment; expansion to additional targets such as PfATP1 and PfPI4K would strengthen the polypharmacological assessment.

PfCRT PNS sensitivity. PfCRT (PF3D7_0709000) has no curated STRING PPI interactions at the 700 confidence threshold used in this study. Consequently, its composite centrality defaults to a uniform weight ($C_j = 1.0$), effectively treating PfCRT engagement as independent of network topology. We report PNS rankings with and without the PfCRT default in Table 8 to allow readers to assess its impact.

Force field heterogeneity. Ligands bound to PfCRT (membrane protein) were parameterised with CGenFF v4.4, while ligands bound to PfDHFR, PfATP4, and PfClpP (soluble proteins) were parameterised with OpenFF 2.2 (Sage). While this choice reflects best practices for each environment, it introduces a systematic heterogeneity that may affect inter-target comparisons.

Monte Carlo force field mismatch. MC sampling used the AMBER14 force field (via OpenMM), while MD simulations used CHARMM36m (via GROMACS). Although both are well-validated for protein-ligand systems, the force field difference introduces a systematic offset in absolute binding free energies. We address this by focusing on relative rankings and convergence metrics rather than absolute values.

Statistical power. The cross-metric correlation analysis is based on $n = 20$ candidates, which provides $\sim 72\,\%$ power to detect moderate correlations (Spearman $\rho = 0.5$) at the Bonferroni-corrected $\alpha = 0.008$. Effect sizes are reported alongside p-values to mitigate this limitation. Future work with larger candidate sets will improve statistical robustness.

PfCRT protonation state. PfCRT functions in the acidic digestive vacuole of *P. falciparum* (pH 5.0–5.4), but docking and MD simulations were performed with protein protonation states assigned at physiological pH 7.4. In companion work (Paper 1), re-protonation of PfCRT at pH 5.2 using PROPKA3 revealed that eight active-site residues exhibited shifted protonation states, leading to a systematic decrease in mean binding affinity of 2.20 kcal mol^{-1} and a Spearman ranking correlation of only $\rho = 0.270$ between pH 7.4 and pH 5.2 scores. While the structural persistence observed during MD simulation (ligand RMSD 1.15 Å, 100 % binding occupancy) suggests that PfCRT-bound candidates maintain binding at acidic pH, the absolute docking scores reported here should be interpreted as upper-bound estimates. Future PfCRT-targeted campaigns should apply vacuolar pH-corrected protonation states throughout the docking and scoring pipeline.

5 Conclusion

We present a resistance-aware computational framework for antimalarial lead validation that integrates production molecular dynamics (40 ns across 4 wild-type complexes) with docking-based resistance resilience scoring across 220 protein–ligand systems and Monte Carlo binding free energy landscape exploration. Three novel scoring metrics are introduced: the Resistance Resilience Score classifies polypharmacological leads by their ability to maintain binding affinity across six African-prevalent resistance mutations; the African Chemical Space Index quantifies the degree to which generated compounds preserve the structural character of African natural products; and the Polypharmacology Network Score weights multi-target engagement by the topological importance of each target in the *P. falciparum* interactome. DiffDock–Vina correlation analysis ($r = 0.327$–0.361 for PfDHFR/PfClpP, negligible for PfCRT/PfATP4; pooled $r = -0.05$) confirms that the two docking methods provide complementary information, justifying the dual-filter consensus strategy. Benjamini–Hochberg correction across 18 activity comparisons yielded zero significant results, confirming statistical parity between generated analogues and foundational natural products. Post-hoc trajectory analysis confirms that two of the four wild-type systems maintain bound ligands (PfCRT 214 and PfATP4 438), while PfClpP (164) and PfDHFR (201) are unbound — demonstrating that MD is a mandatory post-docking filter. The mutant systems have docking-based RRS classifications awaiting full production MD validation. All scripts, trajectories, and scoring metrics are deposited on Zenodo and GitHub to support resistance-aware antimalarial discovery.

Author Contributions

Conceptualization and methodology, M.V.S.T. and S.G.N.E.; software, M.V.S.T. and J.-P.T.N., P.S.; validation, M.V.S.T., J.-P.T.N. and W.F.M.; formal analysis and investigation, M.V.S.T. and S.G.N.E.; resources, S.G.N.E. and W.F.M., P.S.; data curation, M.V.S.T.; writing–original draft preparation, M.V.S.T.; writing–review and editing, M.V.S.T., J.-P.T.N., P.S., W.F.M. and S.G.N.E.; visualization, M.V.S.T.; supervision, S.G.N.E. and W.F.M.; project administration, S.G.N.E. All authors read and agreed to the published version.

Data Availability

All MD trajectories, MM-GBSA outputs, and analysis scripts are deposited on Zenodo (DOI: `https://doi.org/10.5281/zenodo.19608875`, shared with the companion chemical-space dataset). Docking results and compound SMILES are provided as supplementary CSV files, including the full molecular library in SELFIES representation (`all_mol_selfies.csv`, 51 molecules). All code is available at `https://github.com/NanaEngo/Malaria_codesV2` under the MIT licence.

Competing Interests

The authors declare no competing interests.

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